Activation functions are now a search space, not a fixed design choice.

We optimize the distribution of activation functions across hidden neurons in small neural networks.

- Gradient descent trains weights for a fixed layout.
- Simulated Annealing searches over discrete activation layouts.
- Final claims come from retrained multi-seed comparisons.

layout = AF assignment per hidden neuron

6
activation functions

3
official benchmarks

10
planned seeds

The implementation separates continuous weight training from discrete layout search.

Weight training

- Mini-batch gradient descent updates weights and biases.
- The activation layout is fixed during a training run.
- Metrics are collected on train / validation / test splits.

Layout search

- SA proposes local activation-layout changes.
- Candidates are evaluated by loss, not test accuracy.
- Selected layouts are retrained for final comparison.

same model core, different optimization target

Implementation anchor

online_annealing_training_service.py contains the new default SA path.

The refined method judges layout changes before retraining the candidate.

Earlier comparison path

short-retrain

1. **Propose layout**
   Generate a candidate activation assignment.
2. **Fresh model + short training**
   Candidate gets a small training budget before scoring.

Useful as a legacy comparison, but the score mixes layout quality with short-training noise.

Final method

online-delta

1. **Same weights + same batch**
   Evaluate the current network state.
2. **Change AF layout only**
   Measure the direct loss delta before training.

Acceptance is based on loss delta; training happens after acceptance.

One SA step compares loss before and after an activation-layout change.

1 **Evaluate current**
loss_before on current mini-batch

2 **Apply neighbor**
same weights, new activation layout

3 **Evaluate candidate**
candidate_loss_after on same mini-batch

4 **Accept / reject**
delta = after - before; p = exp(-delta / T)

If accepted: keep the candidate layout and train the inherited model on that mini-batch.

If rejected: restore the previous layout and move to the next proposal.

Test data is never used during search

The comparison frame is fixed before interpreting any result.

Difficulty	Benchmark	Topology	Training budget	Purpose
Easy	concentric_circles	2-8-1	100 epochs	sanity check for visible learning
Medium	iris	4-8-3	150 epochs	main Wednesday focus
Hard	crossing_spirals	6-16-16-1	250 epochs	limitation / stress case

Activation set: relu | gelu | sigmoid | tanh | swish | identity

CSV benchmark source: Basics Group repository, imported at commit ccb9253e2ccaf2b1d8861f72db10cdba64def6ba.

Hyperparameters define the evaluation budget, not the conclusion.

Seeds	Learning rate	Batch size	Weight scale
42-51	0.03	32	0.05

SA steps	Start T	Cooling	Policy
200	0.5	geometric 0.92	train accepted

validation loss
search + ranking signal

test metrics
final reporting only

Final claims require mean and standard deviation across seeds.

E1 establishes how strong each single activation function is without SA.

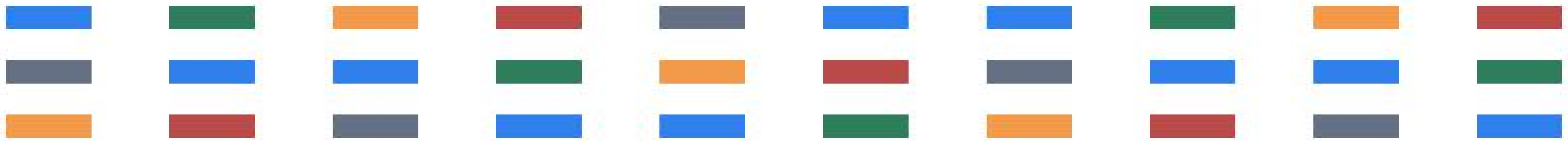
Experiment	Benchmarks	Layouts	Seeds	Main question
E1	all 3	all_relu, all_gelu, all_sigmoid	10	Which single AF works best per benchmark?
		all_tanh, all_swish, all_identity	10	How strong is the non-SA baseline?

Metric	Why it matters
mean_val_loss	primary model-selection metric
std_val_loss	stability across seeds
mean_test_loss	final generalization error
mean_test_accuracy	readable final performance

Without E1, SA results have no fair baseline.

E2 checks whether mixed random layouts are already competitive.

10 random layouts x **10 training seeds** x **3 benchmarks**



Random baseline answers: is SA better than just mixing AFs?

Report both the average random layout and the best random layout; otherwise a single lucky random sample can mislead.

E3-E7 test whether online-delta SA improves five start conditions.

ID	Start layout	SA mode	Final comparison
E3	all_relu	online_delta	start vs best SA vs end SA vs baselines
E4	all_tanh	online_delta	same final evaluation
E5	all_sigmoid	online_delta	can SA escape a weak saturating baseline?
E6	all_identity	online_delta	does SA improve or damage a strong baseline?
E7	random layout	online_delta	does SA improve already mixed layouts?

For Wednesday: run/interpret Iris first. Then extend the same matrix to concentric_circles and crossing_spirals.

Final claims use retrained layouts; inherited models are reported separately.

Early Iris runs show improvement from weak starts, not universal superiority.

Start layout	Search mean val loss	Final SA val loss	Final SA test acc	Strong simple baseline
relu*8	0.5875	0.2533	0.9333	all_identity: 0.2363 val loss
tanh*8	0.5765	0.2469	0.9333	all_identity: 0.2363 val loss
sigmoid*8	0.5891	0.2468	0.9333	all_identity: 0.2363 val loss

Interpretation: online-delta SA improves weak start layouts and finds competitive mixed layouts.

Caveat: all_identity remains slightly stronger by validation loss in these remembered Iris runs. This is why the next deck/result pack needs the full E1-E7 matrix and mean/std plots.

Source: docs/EXPERIMENT_NOTES.md, 8 seeds, online_delta, train policy accepted, max_steps 200.

Ablations come after the core matrix is stable.

E8: old vs new SA

Compare short-retrain against online-delta with the same start layouts, seeds, step budget, and final retraining. The comparison target is final layout quality, not raw search score.

E9-E11: proposal policies

Local-only, balanced, and exploratory operator sampling distributions. This tests how candidate generation affects SA without changing the acceptance rule.

Report question

Does the method improve weak starts reliably, and when do simple baselines remain stronger? This is more defensible than claiming universal superiority.

Do not expand these ablations before E1-E7 are reported cleanly.

The next deliverable is a reproducible result pack, not another GUI feature.

1 **Run E1 and E2**

Single-AF and random-layout baselines on all benchmarks.

2 **Run E3-E7 on Iris**

Focus Wednesday interpretation on the fastest meaningful benchmark.

3 **Extend to all benchmarks**

Repeat the same matrix before comparing hard-case behaviour.

4 **Build report plots**

Mean/std validation loss, test accuracy, SA improvement, and baseline ranking.

5 **Discuss ablations**

Old-vs-new SA and proposal policies after core evidence is stable.

Clean claim boundary: SA can be useful if it improves start layouts under identical final evaluation, even when a simple baseline remains competitive.

Wednesday target: method accepted, experiment matrix locked, first Iris evidence explained.